Software Test Plan (STP) - Hostel Management System

Project: Hostel Management System (HMS)

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Revision History

Version	Date	Contributor	Revision / Contribution
1.0	05-10-2026	G.V.N. Sreeharsha	Initial Test Plan baseline based on the SRS scope and requirements.
1.1	07-10-2026	Chennamraju Vaishnavi	Developed test strategy, test levels and types, environment and test case details; filled the missing SAD reference and extended TC-RPT-01 to cover the payment status report.
1.2	07-10-2026	Devansh Gupta	Reviewed security validation and traceability coverage; added team number and team member details.

1. Introduction

Purpose: This document defines the test plan for the Hostel Management System (HMS). It outlines objectives, scope, strategy, resources, schedule and responsibilities for testing.

Scope: Testing covers authentication and session management, student management, hostel/room/bed management, allocation (allocate, transfer, release), fee payment and receipts, occupancy report, student self-view, and the security and non-functional requirements (performance, data integrity, backup, compatibility).

References: HMS SRS (07-10-2026), HMS SAD (07-10-2026), IEEE Std 829 (Software Test Documentation), OWASP ASVS 4.0, SE Mini-Project Deliverables Part-1.

Definitions: HMS (Hostel Management System), SRS (Software Requirements Specification), RTM (Requirements Traceability Matrix), RBAC (Role Based Access Control), CSRF (Cross Site Request Forgery), TC (Test Case), REQ (functional requirement ID), NFR (non-functional requirement ID).

2. Test Items

- Authentication and session module
- User administration module
- Student management module
- Hostel, room and bed management module
- Allocation module (allocate, transfer, release, history)
- Payment and receipt module
- Reports module (occupancy report)

- Student self-service view
- Audit logging and database (MySQL schema and constraints)

3. Features to be Tested

Features mapped to SRS requirement IDs:

- **REQ-1 to REQ-4:** Login, lockout, logout/session timeout, user administration
- **REQ-5 to REQ-9:** Student registration, duplicate check, search, update, deactivate
- **REQ-10 to REQ-13:** Hostel, room, automatic bed creation, available-bed view
- **REQ-14 to REQ-19:** Allocation, conflict checks, gender rule, transfer, release, history
- **REQ-20 to REQ-22:** Payment recording, receipt, payment status
- **REQ-23 to REQ-24:** Occupancy report, student self-view

- **NFR-1 to NFR-3:** Performance: response time, allocation time, data volume
- **NFR-4, NFR-5:** Safety: daily backup, no permanent deletion
- **NFR-6 to NFR-12:** Security: password hashing, RBAC, own-data access, session, input validation/CSRF, audit log, HTTPS/lockout
- **Quality attributes:** Usability, reliability/data integrity, portability (browser compatibility), maintainability

4. Features Not to be Tested

- Internal code of Django, Python and MySQL (assumed tested by their vendors)
- Browser internals and operating system behaviour
- Third-party library internals (reportlab, mysqlclient)
- Institute network, hosting and power infrastructure
- Online payment gateways, e-mail/SMS (out of scope for HMS v1.0)
- Native mobile applications (not part of v1.0)

5. Test Approach / Strategy

Levels:

- Unit tests (Django TestCase for models, forms and views of each module)
- Integration tests (views ↔ ORM ↔ MySQL, for example allocation transaction and constraints)
- System tests (end-to-end flows from the browser using the test cases in Section 13)
- Acceptance tests (walkthrough of the main flows with the project guide)

Types:

- Functional testing (all REQ items)
- Regression testing (re-run the automated suite after each sprint)
- Performance testing (response time, allocation time, data volume)
- Security testing (see Section 5.1)
- Usability and compatibility testing (browsers, new-user task)

Entry Criteria: SRS approved, build deployed to test environment, test data loaded, test cases reviewed.

Exit Criteria: 100% of planned test cases executed, all High-priority test cases passed, 0 critical or high defects open, every requirement in the RTM covered by at least one passed test case.

5.1 Security Validation

Security validation checks the security objectives SO-1 to SO-4 and requirements NFR-6 to NFR-12 of the SRS:

- **Authentication and lockout:** wrong-password handling, 5-attempt lockout (TC-AUTH-02, TC-AUTH-03)
- **Password protection:** PBKDF2 hashing and password policy (TC-SEC-04)
- **Access control:** role checks on every request, direct-URL access by lower roles, student-only-own-data (TC-SEC-01, TC-SEC-02)
- **Input handling:** SQL injection and script injection attempts in login, search and form fields; CSRF token check (TC-SEC-03, TC-SEC-05)
- **Session and transport security:** 15-minute timeout, HttpOnly/Secure cookies, HTTPS with TLS 1.2 or higher (TC-AUTH-04, TC-SEC-06)
- **Accountability:** audit log of logins and data-changing actions (TC-SEC-07)

- **Automated scans:** OWASP ZAP baseline scan on the running application and Django's "manage.py check --deploy" for deployment settings; findings are logged as defects

6. Test Environment

Hardware: Team laptops (minimum 8 GB RAM) acting as application server and database host; client devices with Chrome, Firefox and Edge, including one mobile phone.

Software: HMS v1.0 build, Python 3.11+, Django 5.x, MySQL 8.0, Windows 10+ or Ubuntu 20.04+.

Tools: Django test runner (unit/integration), Selenium (UI automation), Postman (API checks), JMeter (performance), OWASP ZAP (security scan), GitHub Issues (defect tracking), GitHub (version control).

Test Data: Dummy students (male and female), 2 hostels (1 Boys, 1 Girls) with rooms and beds, users for each role (Admin, Warden, Student), a bulk data set of 5,000 students and 2,000 beds for volume testing. No real student data is used.

7. Test Schedule

Planned milestones (to be aligned with the sprint plan):

- Test plan and test case design: 05-Oct-2026 to 06-Oct-2026
- Test plan submission (Part-1): 07-Oct-2026
- Environment setup and test data preparation: 12-Oct-2026
- Unit and integration testing (alongside each sprint): from Sprint 1
- System test execution: after feature completion, as per sprint plan
- Security and performance testing: after system test execution
- Regression and final test summary: before final project submission

8. Test Deliverables

- Test Plan (this document)
- Test Cases (Section 13) - manual and automated
- Test Scripts (Django tests, Selenium, JMeter plans)
- Test Data
- Test Execution Logs
- Defect Reports (GitHub Issues)
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Chennamraju Vaishnavi	Prepare plan, coordinate execution, prepare summary report
Test Engineer	All team members (own modules)	Design and execute test cases, log defects
Developer	Devansh Gupta	Write unit tests, support defect fixes and triage
Product Owner	G.V.N. Sreeharsha	Approve test results, sign off readiness

Every team member takes part in testing the modules they own.

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build delivery	Ask developers for early builds at the end of every sprint
Test environment downtime (laptop or MySQL failure)	Keep the database script and test data in Git so the environment can be rebuilt quickly
Short time before final submission	Automate high-priority tests early; run the High-priority test cases first
Concurrency defects hard to reproduce	Use scripted parallel requests and database constraints (TC-PERF-02)
Performance targets not met on team laptops	Test with realistic data early and index frequently searched columns

11. Assumptions & Dependencies

- The SRS is approved and its requirement IDs will not change
- Test data will be created before execution
- Each sprint produces a working build that can be tested
- Test laptops have the same software versions as listed in Section 6
- Test cases may be refined once the SAD and implementation are finalised

12. Suspension & Resumption Criteria

Suspend testing if:

- Environment is unavailable for more than 4 hours
- Build is too unstable (blocks more than 30% of test cases)
- A critical defect (for example login failure) blocks all further testing

Resume testing if:

- Blocking defects are fixed and verified
- Environment is stable again

13. Test Case Management & Traceability

The RTM maps every SRS requirement to test cases. In total there are 33 test cases: 21 functional, 8 security, 3 performance and 1 other non-functional, covering all 36 requirements (REQ-1 to REQ-24 and NFR-1 to NFR-12).

13.1 Test Cases

TC ID	Title	Req ID	Type	Steps / Input	Expected Result
TC-AUTH-01	Valid login	REQ-1	Functional	Pre: Warden account exists. 1. Open login page. 2. Enter correct username and password.	Warden dashboard is shown within 2 seconds.
TC-AUTH-02	Invalid login	REQ-2	Functional	1. Enter valid username with wrong password. 2. Enter an unknown username.	Message "Invalid username or password" is shown in both cases; the message does not say which field

					was wrong; user stays logged out.

TC ID	Title	Req ID	Type	Steps / Input	Expected Result
TC-AUTH-03	Account lockout after failed attempts	REQ-2, NFR-12	Security	1. Enter wrong password 5 times in a row. 2. Enter the correct password. 3. Retry after 15 minutes.	Step 2 is rejected with a lockout message; step 3 succeeds.
TC-AUTH-04	Logout and session timeout	REQ-3, NFR-9	Functional	1. Click Logout, then press browser Back. 2. Log in, stay idle for 15 minutes, then click any link.	After logout, protected pages redirect to login. After 15 idle minutes, session ends and login is required.
TC-USR-01	Admin manages user accounts	REQ-4	Functional	1. Admin creates a Warden account. 2. Create another account with the same username. 3. Deactivate the Warden and try to log in.	Account created with role; duplicate username rejected; deactivated user cannot log in.
TC-STU-01	Register student (valid data)	REQ-5	Functional	1. Warden opens Add Student. 2. Fill all mandatory fields from Appendix B with valid data and submit.	Student saved; success message shown; student appears in student list.
TC-STU-02	Register student (invalid / duplicate)	REQ-5, REQ-6	Functional	1. Submit with a mandatory field empty. 2. Submit with an existing Student ID.	Step 1: field-level error, nothing saved. Step 2: "Student ID already registered" shown, no second record.
TC-STU-03	Search students	REQ-7	Functional	Search by (a) full Student ID, (b) partial name, (c) hostel.	Correct matching students are listed for each search; no match shows an empty-result message.
TC-STU-04	Update student details	REQ-8	Functional	1. Edit mobile number and save. 2. Try to edit Student ID.	Mobile number updated; Student ID field is read-only / change rejected.
TC-STU-05	Deactivate student	REQ-9, NFR-5	Functional	1. Deactivate a student with an active allocation. 2. Deactivate a student with no allocation.	Step 1 refused and allocated bed shown. Step 2 marks student Inactive; record still exists in the database.
TC-HST-01	Add hostel and room	REQ-10, REQ-11	Functional	1. Admin adds a Boys hostel. 2. Add room 101 (capacity 3). 3. Add room 101 again in the same hostel. 4. Add room with capacity 5.	Hostel and room saved; duplicate room number rejected; capacity 5 rejected (allowed range 1 to 4).
TC-HST-02	Bed creation and availability list	REQ-12, REQ-13	Functional	1. After TC-HST-01, open Available Beds. 2. Filter by hostel, floor, room type. 3. Allocate one bed and refresh.	Beds 101-A, 101-B, 101-C exist as Available; filters work; allocated bed disappears from the list.
TC-ALC-01	Allocate available bed	REQ-14	Functional	Select an active male student and an available bed in the Boys hostel; confirm.	Allocation saved with date; bed status becomes Occupied;

					confirmation shown.
TC-ALC-02	Block allocation of occupied bed	REQ-15	Functional	Try to allocate an already occupied bed to another student.	Request rejected with reason; no second active allocation exists for the bed.
TC-ALC-03	Block second allocation for a student	REQ-15	Functional	Try to allocate a second bed to a student who already has an active allocation.	Request rejected with reason; original allocation unchanged.
TC-ALC-04	Gender and hostel type mismatch	REQ-16	Functional	Try to allocate a female student to a bed in a Boys hostel.	Allocation rejected with a gender mismatch message.
TC-ALC-05	Transfer student	REQ-17	Functional	1. Transfer an allocated student to another available bed. 2. Force a failure while saving the new bed (for example, DB error in test) and retry.	Step 1: old bed Available, new bed Occupied. Step 2: transaction rolled back, old allocation still active (atomic).
TC-ALC-06	Release allocation and history	REQ-18, REQ-19	Functional	1. Release an allocation with a vacate date. 2. Open the student's allocation history.	Bed becomes Available; vacate date stored; history lists all past and current allocations in date order.

TC ID	Title	Req ID	Type	Steps / Input	Expected Result
TC-PAY-01	Record payment and receipt	REQ-20, REQ-21	Functional	1. Record payment (amount 5000, UPI, date, reference). 2. Record a payment with amount 0 and one with negative amount. 3. Download the receipt.	Step 1 saved with a unique sequential receipt number. Step 2 rejected. Step 3 gives a PDF receipt with correct details.
TC-PAY-02	Payment status	REQ-22	Functional	Record no payment, then partial payment, then full payment for a student.	Status shows Pending, then Partial, then Paid.
TC-RPT-01	Occupancy report	REQ-23	Functional	Open the occupancy report and the payment status report for a hostel with known counts; export CSV for both.	Total, occupied, available beds and occupancy % match database counts; CSV downloads with same values.
TC-SEC-01	Role-based access control	REQ-4, NFR-7	Security	1. Log in as Warden and open the staff-account and add-hostel URLs directly. 2. Log in as Student and open student-list and allocation URLs directly.	Every request is blocked with HTTP 403 "Access denied"; no data is changed.
TC-SEC-02	Student sees only own data	REQ-24, NFR-8	Security	1. Student A opens My Allocation. 2. Student A edits the ID in the URL to Student B's.	Step 1 shows only A's data. Step 2 returns 403/404 and shows no data of B.
TC-SEC-03	SQL injection and script injection	NFR-10	Security	1. Enter ' OR '1'='1 in username and in the search box. 2. Enter <script>alert(1)</ script> as student name.	No login bypass; no SQL error is shown; the script is stored/displayed as plain text and does not run.
TC-SEC-04	Password storage and policy	NFR-6	Security	1. Inspect the password column in the database. 2. Create a user with password "abc".	Password is stored as a PBKDF2 hash, not plain text. Weak password rejected (min 8 characters with a letter and a digit).
TC-SEC-05	CSRF protection	NFR-10	Security	Submit a state-changing POST (for example add student) without a valid CSRF token.	Request is rejected (HTTP 403) and no data changes.
TC-SEC-06	HTTPS and secure cookies	NFR-12, NFR-9	Security	1. Open the site over http://. 2. Run a TLS scan. 3. Inspect session cookie flags.	HTTP redirects to HTTPS; only TLS 1.2 or higher accepted; session cookie has HttpOnly and Secure flags.
TC-SEC-07	Audit log	NFR-11	Security	Perform login, failed login, student add, allocation, release and payment; open the audit log.	Each action has a log entry with user, action, record and timestamp.
TC-PERF-01	Response time under load	NFR-1	Performan ce	JMeter: 50 concurrent users for 10 minutes doing login, student search and available-beds view.	At least 95% of requests complete within 2 seconds; no errors.
TC-PERF-02	Allocation time and concurrency	NFR-2	Performan ce	1. Allocate beds 20 times and measure time. 2. Send two	Each allocation takes under 3 seconds. Only one concurrent request

				simultaneous requests for the same bed.	succeeds; the bed has exactly one active allocation.
TC-PERF-03	Data volume	NFR-3	Performan ce	Load 5,000 students and 2,000 beds; repeat TC-PERF-01 and student search.	Response times still meet NFR-1; search returns within 2 seconds.
TC-OPS-01	Backup and restore	NFR-4	Functional	Check that the daily backup job ran; restore the latest backup into a test database.	Backup file exists for the last 24 hours; restore succeeds and data is complete up to the backup time.
TC-QUAL-01	Browser compatibility and usability	Quality attributes (Portability , Usability)	Non-functional	1. Run the main flows on latest Chrome, Firefox and Edge. 2. Ask a new warden to register a student and allocate a bed using the one-page guide.	All flows work on the three browsers. The task is completed within 5 minutes.

13.2 Requirements Traceability Matrix

Req ID	Requirement	Test Case ID(s)
REQ-1	Login using ID and password, role dashboard	TC-AUTH-01
REQ-2	Invalid login message and account lockout	TC-AUTH-02, TC-AUTH-03
REQ-3	Logout and session timeout	TC-AUTH-04
REQ-4	Admin creates / deactivates users and assigns roles	TC-USR-01, TC-SEC-01
REQ-5	Register student with mandatory fields	TC-STU-01, TC-STU-02
REQ-6	Reject duplicate Student ID	TC-STU-02
REQ-7	Search students	TC-STU-03
REQ-8	Update student details	TC-STU-04
REQ-9	Deactivate student only if no active allocation	TC-STU-05
REQ-10	Admin adds / edits hostel	TC-HST-01
REQ-11	Admin adds room with capacity	TC-HST-01
REQ-12	Auto-create beds for a room	TC-HST-02
REQ-13	Show available beds with filters	TC-HST-02
REQ-14	Allocate available bed to student	TC-ALC-01
REQ-15	Block occupied bed / second allocation	TC-ALC-02, TC-ALC-03
REQ-16	Gender and hostel type must match	TC-ALC-04
REQ-17	Transfer student to another bed	TC-ALC-05
REQ-18	Release allocation	TC-ALC-06
REQ-19	Allocation history	TC-ALC-06
REQ-20	Record payment	TC-PAY-01
REQ-21	Generate receipt	TC-PAY-01
REQ-22	Show payment status	TC-PAY-02
REQ-23	Occupancy report	TC-RPT-01
REQ-24	Student views own data	TC-SEC-02
NFR-1	95% of pages within 2 s for 50 users	TC-PERF-01
NFR-2	Allocation within 3 s	TC-PERF-02
NFR-3	5,000 students and 2,000 beds	TC-PERF-03
NFR-4	Daily database backup	TC-OPS-01
NFR-5	No permanent deletion of records	TC-STU-05
NFR-6	Password hashing and policy	TC-SEC-04
NFR-7	Role check on every request	TC-SEC-01
NFR-8	Student sees only own data	TC-SEC-02
NFR-9	Session expiry and secure cookies	TC-AUTH-04, TC-SEC-06
NFR-10	Input validation, CSRF, SQL injection	TC-SEC-03, TC-SEC-05
NFR-11	Audit logging	TC-SEC-07
NFR-12	HTTPS and brute-force lockout	TC-AUTH-03, TC-SEC-06

14. Test Metrics & Reporting

Metrics collected:

- % of test cases executed
- % passed / failed
- Defect density (defects per module)
- Defect aging (days open)
- Requirement coverage (requirements with at least one passed test case)

Reports:

- Execution status after every sprint
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead	Chennamraju Vaishnavi	
Dev Lead	Devansh Gupta	
Product Owner	G.V.N. Sreeharsha	